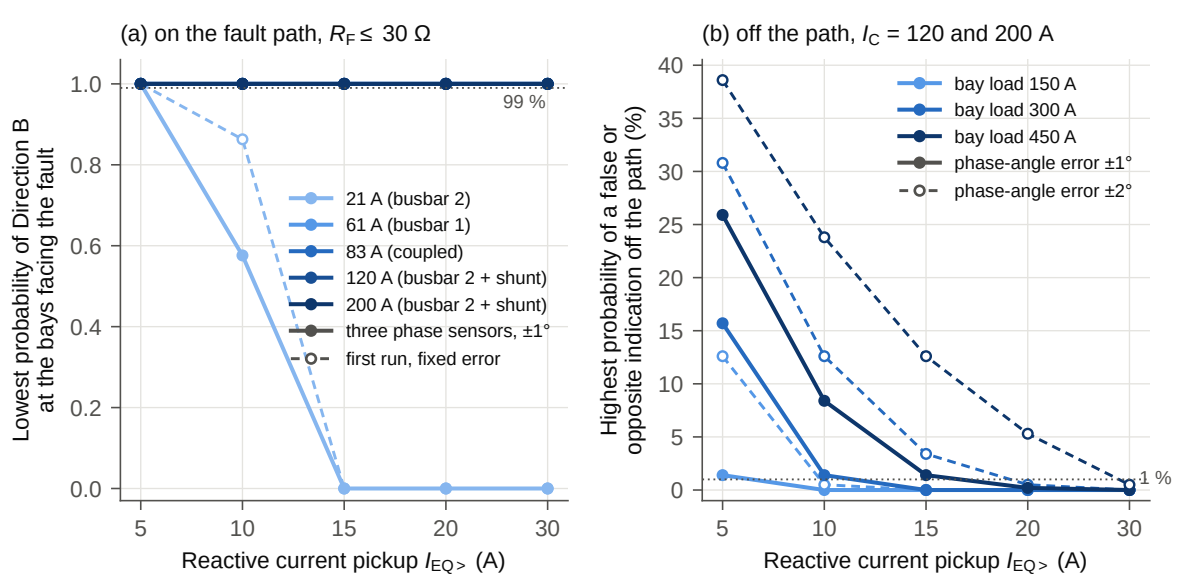
(a) on the fault path, $R_F \leq 30\ \Omega$
Lowest probability of Direction B at the bays facing the fault
1.0
0.8
0.6
0.4
0.2
0.0
99 %
21 A (busbar 2)
61 A (busbar 1)
83 A (coupled)
120 A (busbar 2 + shunt)
200 A (busbar 2 + shunt)
three phase sensors, ±1°
first run, fixed error
5
10
15
20
30
Reactive current pickup $I_{EQ>}$ (A)
(b) off the path, I_C = 120 and 200 A
Highest probability of a false or opposite indication off the path (%)
40
35
30
25
20
15
10
5
0
bay load 150 A
bay load 300 A
bay load 450 A
phase-angle error ±1°
phase-angle error ±2°
1 %
5
10
15
20
30
Reactive current pickup $I_{EQ>}$ (A)